

Theorem (Jensen's Inequality). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be convex; that is,*

$$\forall x, y \in \mathbb{R}, \forall t \in [0, 1] : f(tx + (1-t)y) \leq tf(x) + (1-t)f(y). \quad (C)$$

Let $x_1, \dots, x_n \in \mathbb{R}$ and let weights $\lambda_1, \dots, \lambda_n$ satisfy $\lambda_i \geq 0$ for all i and $\sum_{i=1}^n \lambda_i = 1$. Then

$$f\left(\sum_{i=1}^n \lambda_i x_i\right) \leq \sum_{i=1}^n \lambda_i f(x_i). \quad (J)$$

1. Degenerate cases If $\lambda_k = 1$ for some index k (hence all other $\lambda_i = 0$), then

$$\sum_{i=1}^n \lambda_i x_i = x_k, \quad \sum_{i=1}^n \lambda_i f(x_i) = f(x_k),$$

so (J) reduces to the identity $f(x_k) \leq f(x_k)$, which holds with equality. If any weight equals zero, the corresponding term disappears from both sides of (J); therefore such indices may be omitted without affecting the statement.

2. Base case $n = 2$ Let $\lambda \in [0, 1]$ and set $\lambda_1 = \lambda$, $\lambda_2 = 1 - \lambda$. Applying convexity (C) with $t = \lambda$, $x = x_1$, $y = x_2$ yields

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2),$$

which is exactly (J) for $n = 2$.

3. Inductive hypothesis Assume that for some integer $m \geq 2$ the inequality holds for every collection of m points with non-negative weights summing to one:

$$f\left(\sum_{i=1}^m \lambda_i x_i\right) \leq \sum_{i=1}^m \lambda_i f(x_i) \quad (\lambda_i \geq 0, \sum_{i=1}^m \lambda_i = 1). \quad (IH)$$

4. Inductive step $m \rightarrow m+1$ Let $\lambda_1, \dots, \lambda_{m+1} \geq 0$ with $\sum_{i=1}^{m+1} \lambda_i = 1$ and set $\mu := \lambda_{m+1}$.

If $\mu = 1$ we are back to the degenerate case. Otherwise $0 \leq \mu < 1$. Define normalized weights for the first m indices:

$$\alpha_i := \frac{\lambda_i}{1 - \mu}, \quad i = 1, \dots, m.$$

Since $\sum_{i=1}^m \lambda_i = 1 - \mu$, we have

$$\alpha_i \geq 0, \quad \sum_{i=1}^m \alpha_i = 1. \quad (4.1)$$

Now rewrite the full convex combination as a two-point convex combination:

$$\sum_{i=1}^{m+1} \lambda_i x_i = (1 - \mu) \left(\sum_{i=1}^m \alpha_i x_i \right) + \mu x_{m+1}. \quad (4.2)$$

Applying convexity (C) with $t = 1 - \mu$ to the points $y := \sum_{i=1}^m \alpha_i x_i$ and $x := x_{m+1}$ gives

$$f\left(\sum_{i=1}^{m+1} \lambda_i x_i\right) \leq (1 - \mu) f\left(\sum_{i=1}^m \alpha_i x_i\right) + \mu f(x_{m+1}). \quad (4.3)$$

By the inductive hypothesis (IH) applied to the m points $\{x_i\}_{i=1}^m$ with weights $\{\alpha_i\}$,

$$f\left(\sum_{i=1}^m \alpha_i x_i\right) \leq \sum_{i=1}^m \alpha_i f(x_i). \quad (4.4)$$

Inserting (4.4) into (4.3) yields

$$f\left(\sum_{i=1}^{m+1} \lambda_i x_i\right) \leq (1 - \mu) \sum_{i=1}^m \alpha_i f(x_i) + \mu f(x_{m+1}). \quad (4.5)$$

Since $\alpha_i = \lambda_i / (1 - \mu)$, we have $(1 - \mu)\alpha_i = \lambda_i$ for $i = 1, \dots, m$; thus the right-hand side of (4.5) becomes

$$\sum_{i=1}^m \lambda_i f(x_i) + \lambda_{m+1} f(x_{m+1}) = \sum_{i=1}^{m+1} \lambda_i f(x_i),$$

which is precisely (J) for $m + 1$ points. Hence the statement holds for all $n \geq 1$ by induction.

5. Equality case Equality in (J) can occur only if every inequality used above is an equality. In the base case $n = 2$ this happens when $\lambda \in \{0, 1\}$, when $x_1 = x_2$, or more generally when f is affine on the segment joining x_1 and x_2 . In the inductive step, equality in (4.3) requires either $\mu \in \{0, 1\}$ or $\sum_{i=1}^m \alpha_i x_i = x_{m+1}$; equality in (4.4) forces, by the induction hypothesis, that all points with positive α_i coincide (or that f is affine on their convex hull). Consequently, a simple sufficient condition for equality is that ****all points with positive weight are identical****; if f is strictly convex, this condition is also necessary. □